

Project Description, BMEN 415

Final Project

Due Date April 14, 11:59 PM (Last day of classes)

In the BMEN 415 Design Project, your team will investigate developing three machine learning models (a regression model, a non-neural network classifier, and a neural network classifier) on an open-source dataset containing vital signs from sensors, lab values, and demographics. The focus is not on model accuracy but on decision-making through model design and verification.

As a team, you will develop a regression, non-neural-network, and neural-network classification models, focusing on the training regime (including training, validation, and test data portions) and feature selection, with a rationale tied to physiology.

As individuals, you will propose design decisions and conduct verification activities to investigate and evaluate design decisions under the baseline models and proposed alternatives. Members of the team cannot make the same design decision verifications, and each student's design decision verifications must be agreed upon by the team (to ensure they are unique within the team and do not clash with the team baseline) and approved by the TA to ensure that it is non-trivial, testable, and scoped. Unapproved decisions/experiments will not be graded.

Team Compositions

Teams are self-selected, with 4 students per team. Please note that instructors can assign or remove student(s) from self-selected teams at any time to ensure team sizes of 4 students are achieved.

Groups should endeavour to work consistently throughout the term, dividing work so that all members are involved in all types of work. Groups should attempt to resolve any conflicts they might have within their group, but the lead instructor(s) can also be approached if a third party is needed. Individual grades can be reduced relative to the group mark as needed. This process is initiated by individuals within the group.

Scenario

Sepsis is a critical condition that occurs when the body's reaction to infection leads to tissue damage, organ failure, or death (Singer et al., 2016). Prompt detection and administration of antibiotics are essential for better sepsis outcomes, as each hour of delay can increase mortality risk by approximately 4-8% (Seymour et al., 2017). The aim of this final project is to investigate the design decisions involved in developing a functional, open-source algorithm* that, using only clinical data, can automatically determine a patient's Mean Arterial Pressure (MAP) and sepsis risk, providing a positive

- Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*. 2016 Feb 23;315(8):801-10. doi: 10.1001/jama.2016.0287. PMID: 26903338; PMCID: PMC4968574.
- Seymour CW, Gesten F, Prescott HC, et al. Time to Treatment and Mortality during Mandated Emergency Care for Sepsis. *N Engl J Med*. 2017 Jun 8;376(23):2235-2244. doi: 10.1056/NEJMoa1703058. Epub 2017 May 21. PMID: 28528569; PMCID: PMC5538258.

or negative prediction of sepsis. This project is based on the PhysioNet/Computing in Cardiology Challenge 2019. Further details on the Cardiology Challenge are available at <https://physionet.org/content/challenge-2019/1.0.0/>.

* Focus is *not* on model accuracy, but rather on decision making through model design, trade-offs, and verification.

Dataset and Features

The dataset is available at <https://physionet.org/content/challenge-2019/1.0.0/>. 2026-02-26 Update: Data is available for download at the bottom of the page under “Files” (the “Accessing the Data” hyperlink is broken).

Files

Total uncompressed size: 327.3 KB.

Access the files

- Download the files using your terminal: `wget -r -N -c -np https://physionet.org/files/challenge-2019/1.0.0/`
- Download the files using AWS command line tools: `aws s3 sync --no-sign-request s3://physionet-open/challenge-2019/1.0.0/ DESTINATION`

Folder Navigation: <base>		
Name	Size	Modified
hackathon		
img		
papers		
sources		
training		
LICENSE.txt	25.2 KB	2019-08-05
SHA256SUMS.txt	264 B	2019-08-05
physionet_challenge_2019_ccm_manuscript.pdf	1.5 MB	2019-11-04
utility_nonsepsis_diagram.svg	138.1 KB	2019-07-23
utility_sepsis_diagram.svg	163.7 KB	2019-07-23

Features in yellow are outputs that cannot be used as inputs. Features marked with a cross cannot be used in this project because they are too closely related to the outputs.

Vital signs (columns 1-8)	
HR	Heart rate (beats per minute)
O2Sat	Pulse oximetry (%)
Temp	Temperature (Deg C)
SBP	Systolic BP (mm Hg)
MAP	Mean arterial pressure (mm Hg)
DBP	Diastolic BP (mm Hg)
Resp	Respiration rate (breaths per minute)
EtCO2	End tidal carbon dioxide (mm Hg)
Laboratory values (columns 9-34)	
BaseExcess	Measure of excess bicarbonate (mmol/L)
HCO3	Bicarbonate (mmol/L)

FiO2	Fraction of inspired oxygen (%)
pH	N/A
PaCO2	Partial pressure of carbon dioxide from arterial blood (mm Hg)
SaO2	Oxygen saturation from arterial blood (%)
AST	Aspartate transaminase (IU/L)
BUN	Blood urea nitrogen (mg/dL)
Alkalinephos	Alkaline phosphatase (IU/L)
Calcium	(mg/dL)
Chloride	(mmol/L)
Creatinine	(mg/dL)
Bilirubin_direct	Bilirubin direct (mg/dL)
Glucose	Serum glucose (mg/dL)
Lactate	Lactic acid (mg/dL)
Magnesium	(mmol/dL)
Phosphate	(mg/dL)
Potassium	(mmol/L)
Bilirubin_total	Total bilirubin (mg/dL)
TroponinI	Troponin I (ng/mL)
Hct	Hematocrit (%)
Hgb	Hemoglobin (g/dL)
PTT	partial thromboplastin time (seconds)
WBC	Leukocyte count (count*10 ³ /μL)
Fibrinogen	(mg/dL)
Platelets	(count*10 ³ /μL)
Demographics (columns 35-40)	
Age	Years (100 for patients 90 or above)
Gender	Female (0) or Male (1)
Unit1	Administrative identifier for ICU unit (MICU)
Unit2	Administrative identifier for ICU unit (SICU)
HospAdmTime	Hours between hospital admit and ICU admit
ICULOS	ICU length-of-stay (hours since ICU admit)
Outcome (column 41)	
SepsisLabel	For sepsis patients, SepsisLabel is 1 if SepsisLabel is 0.

Download the dataset and load it in Python to perform some exploratory data analysis and plots. I would recommend moving the data into a comma-separated values (CSV) file, so you can commit it to the team repository (e.g., OneDrive) and use simple commands to open and visualize it.

Resources: Teams are encouraged to review publications from the Cardiology Challenge 2019 (<https://physionet.org/files/challenge-2019/1.0.0/papers/index.html>) to investigate the dataset. However, you must cite any papers you get inspiration from and cannot just copy and paste (e.g., you can recreate figures in papers that describe the data in a way you like).

Design Review #1: Exploratory Data Visualization

During your Week 7 Design Studio, your team will present your preliminary exploratory data visualization (no machine learning component). Keep in mind that we will train this data on several models each, so don't select features that are impractical to work with. For example, lots of missing data, overly small or overly large, and no relation between the features and the targets. Initial feature sets should have a rationale tied to physiology.

The design review serves as an important gateway to advancing your project. Teams that have not received approval and/or are having difficulty with preliminary explanatory data visualization will have an opportunity to regroup for a second design review during Week 8. There will be a dedicated Dropbox for exploratory data analysis that closes the night before your design review, giving your TA a chance to review it beforehand.

Your team is required to document the outcome of your design review, including the review date, attendee names, the design stage, and a summary of action items to follow.

Associated Deliverables: Exploratory Data Visualization of the PhysioNet/Computing in Cardiology Challenge 2019 dataset

- Preliminary Results Date: Design Review #1 (Group Submission)
- Final Submission Date: Tuesday, April 14

Models

Each team of 3-4 students will conduct: 1) a regression case study (predict MAP), 2) a classification case study (predict Sepsis) using a non-neural network, and 3) a classification case study (predict Sepsis) using a neural network, in which you will develop and evaluate multiple models and compare their performance. You can only use the models covered in the course. Each team will share a fixed seed for their simulations to allow reproducibility.

The team will work together to build single-regression, non-neural-network, and neural-network classification models, focusing on the training regime (including training, validation, and test data portions) and feature set selection, with a rationale tied to physiology. Training/Validation/Test sets must be split by patient ID (not by rows). Otherwise, the same patient's later rows could leak into validation/test sets. Each team member will be responsible for investigating their own design decisions and verification plans. The model trained as a team will serve as the baseline for design verification.

Resources: Teams are encouraged to review publications from the Cardiology Challenge 2019 (<https://physionet.org/files/challenge-2019/1.0.0/papers/index.html>) to justify your design choice (the focus is *not* on model accuracy). However, you must cite any papers you get inspiration from and cannot just copy and paste a full model implementation.

Associated Deliverables: Regression, Non-Neural Network Classification Model, and Neural Network Classification Model with training regime and feature set selection.

- Preliminary Results Date: None (Focus is on model design, *not* model performance)
- Final Submission Date: Tuesday, April 14

Design Decisions

The focus is *not* on model accuracy, but rather on decision-making through evaluating alternatives and trade-offs among feature sets and machine learning algorithms, balancing performance, robustness, interpretability, and maintainability. Each team member will be responsible for three design decisions and their verification (one per team member per model). Members of the team cannot make the same design decision, and each student's design decisions must be agreed upon by the team (to ensure they are unique within the team and do not clash with the team baseline) and approved by the TA to ensure that it is non-trivial, testable, and scoped. Unapproved decisions/experiments will not be graded.

A non-exhaustive list of design decisions includes:

- Imputation of missing data
- Class imbalance
- Regression Models
- Classification Models
- Model Hyperparameter Configurations
- Training/Evaluation/Test Metrics
- Constraints

Keep in mind:

- Visualization: How will you present your results?
- Reproducibility: Use fixed seeds and a one-command run script to train both models and generate metrics

Resources: Teams are encouraged to review publications from the Cardiology Challenge 2019 (<https://physionet.org/files/challenge-2019/1.0.0/papers/index.html>) to justify your design choice (the focus is *not* on model accuracy). However, you must cite any papers you get inspiration from and cannot just copy and paste a full model implementation.

Associated Deliverables: Each student has one unique design decision per model (a total of three). Members of the team cannot investigate the same design decision.

- Preliminary Results Date: Design Review #2 (Individual Submission) for one design decision
- Final Submission Date: Tuesday, April 14

Design Verification

Design verification tests that the models work as intended. As part of your verification plan, you are required to formulate verification activities to evaluate design decisions. It is not required that all design decisions be verified. In fact, we recommended limiting the number of design decisions to 1 per team member per model (so three design verifications per team member). Part of the design verification is to identify and describe potential alternative design decisions compared to the baseline (or “do nothing”) model. For example, if your design decision is missing data, the “do nothing” is to delete all rows with missing data (so there is no missing data to deal with), and the alternative is to use imputation. Each student proposes one alternative to the team baseline (“do nothing”) for your design verification. As a result, each team member is responsible for three design decision verifications (regression, non-neural network classification, and neural network classification). Members of the team cannot make the same design decision verifications, and each student’s design decision verifications must be agreed upon by the team (to ensure they are unique within the team and do not clash with the team baseline) and approved by the TA to ensure that it is non-trivial, testable, and scoped. Unapproved decisions/experiments will not be graded.

Verification tests that the model, under the team baseline and the proposed alternative, meets the pre-declared acceptance threshold on the held-out set. Evidence is a single figure/table. Design verification must be restricted to measurable, more technical specifications that quantify function or performance through testing or analysis. Your verification plan must include hypotheses about how the models will perform that are validated before you run your experiments, an acceptance threshold, and a single figure/table you will produce as evidence. Once you have established a verification plan, you are required to conduct your verification testing, analyze and explain the outcomes, and provide a reflection on your hypotheses.

Design Review #2: Design Verification Plan Feedback

During your Week 12 Design Studio, your team will present your design verification plan (proposed hypothesis → acceptance threshold → figure/table for one design decision/model) for approval in a design review. Teams that have not received approval and/or are having difficulty developing a verification plan will have an opportunity to regroup for a second design review during Week 13.

There will be a dedicated Dropbox for the design verification plan that closes the night before your design review, giving your TA a chance to review it beforehand.

Your team is required to document the outcome of your design review, including the review date, attendee names, the design stage, and a summary of action items to follow.

Associated Deliverables: Each student for each design decision will have a proposed hypothesis, acceptance threshold, and figure/table as evidence for each model (so three hypotheses, three acceptance thresholds, and three figures/tables as evidence)

- Preliminary Results Date: Design Review #2 (Individual Submission) for one design verification plan
- Final Submission Date: Tuesday, April 14

Final Submission

Each group will provide a submission that includes code, Exploratory Data Analysis, Baseline Models Overview, Design Decisions and Justification, and Design Verification Plan, Results, Analysis, and Reflection for each team member. For submission, there is no limit on length, but excessive writing is not preferred. There will be a group grade and an individual grade for the project.

Change Log

- 2026-02-26: Specify the location of data and broken hyperlink